day-06 Calculating Jacobians and hessians by hand

Mathmatial Assignment

Problem 1: The jacobin matrix

Consider a layer in a NNs that maps a 2D input vector $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ to a 2D output vector $y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ using the following non linear activation function:

$$y_1 = x_1^2 x_2$$
$$y_2 = 5x_1 + \sin(x_2)$$

Task A: Calci the four first-order partial derivatives required for the jacobin: $\frac{\partial y_1}{\partial x_1}, \frac{\partial y_2}{\partial x_2}, \frac{\partial y_1}{x_2}, \frac{\partial y_2}{\partial x_1}$

Task B: Construct the 2x2 jacobin matrix $J(x)$.

Task C: Evaluate the exact jacobin matrix at the point $(x_1 = 2, x_2 = 0)$.

Solution:

Task A: $\frac{\partial y_1}{\partial x_1} = 2x_1 x_2$ $\qquad$ $\frac{\partial y_1}{\partial x_2} = x_1^2$

$$\frac{\partial y_2}{\partial x_1} = 5 \qquad \frac{\partial y_2}{\partial x_2} = \cos x_2$$

Task B:

$$J(x) = \begin{bmatrix} 2x_1x_2 & x_1^2 \\ 5 & \cos(x_2) \end{bmatrix}$$

Task C: at $x_1 = 2$, $x_2 = 0$

$$J(2,0) = \begin{bmatrix} 0 & 4 \\ 5 & 1 \end{bmatrix}$$

problem 2: The hessian matrix

Now, consider a scalar valued loss function L depent on two weights, w_1 and w_2:

$$L(w_1, w_2) = w_1^3 + 2w_1w_2^2 - 4w_2$$

Task A: (The gradient) calsi the first order partial derivatives $\frac{\partial L}{\partial w_1}$, $\frac{\partial L}{\partial w_2}$. Peak them into gradient vector ∇L.

Task B: Calci the four second order partial derivatives:

- Diff $\frac{\partial L}{\partial w_1}$ w.r.to w_1 to find $\frac{\partial^2 L}{\partial w_1^2}$
- " " " find $\frac{\partial^2 L}{\partial w_1 \partial w_2}$
- " $\frac{\partial L}{\partial w_2}$ " " $\frac{\partial^2 L}{\partial w_2 \partial w_1}$
- " " " " $\frac{\partial^2 L}{\partial w_2^2}$

Solution:

Task A:

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix} = \begin{bmatrix} 3w_1^2 + 2w_2^2 \\ 4w_1w_2 - 4 \end{bmatrix}$$

Task B:

$$\frac{\partial^2}{\partial w_1^2} = \frac{\partial}{\partial w_1}\left(3w_1^2 + 2w_2^2\right) = 6w_1$$

$$\frac{\partial^2 L}{\partial w_1 \partial w_2} = \frac{\partial}{\partial w_2}\left(3w_1^2 + 2w_2^2\right) = 4w_2$$

$$\frac{\partial^2 L}{\partial w_2 \partial w_1} = 4w_2$$

$$\frac{\partial^2 L}{\partial w_2^2} = \frac{\partial}{\partial w_2}\left(4w_1w_2 - 4\right) = 4w_1$$

Task C: The mix partials match ($4w_2 = 4w_2$).

The matrix is perfectly symmetric.

$$H = \begin{bmatrix} 6w_1 & 4w_2 \\ 4w_2 & 4w_1 \end{bmatrix}$$

problem 3: Curvature classification

Suppose your optimizer arrives at the critical point ($w_1 = 1$, $w_2 = 0$).

Task A: Evalate your Hessian matrix H from problem 2 at the specific coordinate ($w_1 = 1$, $w_2 = 0$)

Task B: Find the eigenvalues of this evaluated hessian matrix

Task C: Based on the signs of the eigenvalues, is this critical point a local minimum, local max, or saddle point?

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Solution:

Task A: $w_1 = 1$, $w_2 = 0$

$$H(1,0) = \begin{bmatrix} 6(1) & 4(0) \\ 4(0) & 4(1) \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 4 \end{bmatrix}$$

Task B: the matrix is diagonal, the eigenvalues are simply the entries on the main diagonal $\lambda_1 = 6$, $\lambda_2 = 4$.

Task C: Since the Both eigen value are positive, the matrix is positive Defint. this critical point is a Local minimum.